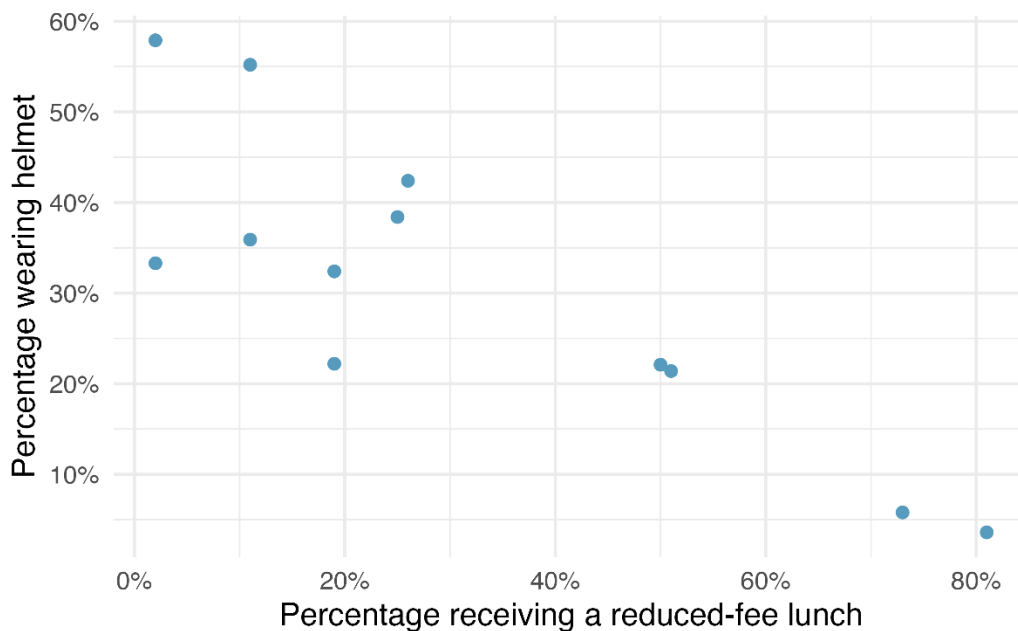


MATH 213 – Homework 10

1. The scatterplot shows the relationship between socioeconomic status measured as the percentage of children in a neighborhood receiving reduced-fee lunches at school (lunch) and the percentage of bike riders in the neighborhood wearing helmets (helmet). The average percentage of children receiving reduced-fee lunches is 30.833% with a standard deviation of 26.724% and the average percentage of bike riders wearing helmets is 30.883% with a standard deviation of 16.948%.



- a. What are the cases for this study? (Hint: What does each dot in the scatterplot represent?)

The cases are the neighborhoods being observed.

- b. What are the variables that are recorded about the cases? Classify each as numerical or categorical.

Lunch (numerical)

Helmet (numerical)

- c. Write a description of the relationship in the scatterplot. Remember the THREE key features that must be addressed, speak in context, and remember GEE.

This scatterplot shows the relationship between the percentage of children in a neighborhood receiving reduced-fee lunches and the percentage of bike riders wearing helmets. Generally, there is a negative association between these variables: neighborhoods with a lower percentage of children receiving reduced-fee lunches tend to have a higher percentage of bike riders wearing helmets, while neighborhoods with a higher percentage of reduced-fee lunches tend to have fewer helmeted riders

Additionally the general trend of the graph is linear and decreasing for example, one neighborhood where about 10% of children receive reduced-fee lunches has around 33 - 37% of bike riders wearing helmets, while another neighborhood with 50% of children receiving reduced-fee lunches has only about 20% of riders wearing helmets.

The relationship is moderately strong, meaning there is some variation in helmet use even among neighborhoods with similar reduced-fee lunch percentages. However, there is an exception to the general trend. For instance, one neighborhood with a relatively low percentage of reduced-fee lunches (under 20%) has a lower-than-expected helmet use (around 57%).

2. Back on Homework 6, you considered a dataset from the 2017 Cherry Blossom Run. The full codebook is at <https://www.openintro.org/data/index.php?data=run17>.

I filtered the dataset to examine the data from female contestants only. Then, I created the following model in R using “lm”.

```
```{r}
fit3 <- lm(pace_sec ~ event,
 data = run_females)
summary(fit3)
```
```

Call:
lm(formula = pace_sec ~ event, data = run_females)

Residuals:

| Min | 1Q | Median | 3Q | Max |
|---------|--------|--------|-------|--------|
| -373.54 | -73.91 | -5.91 | 67.62 | 976.46 |

Coefficients:

| | Estimate | Std. Error | t value | Pr(> t) |
|-------------|----------|------------|---------|------------|
| (Intercept) | 610.910 | 1.086 | 562.57 | <2e-16 *** |
| event5K | 171.628 | 2.820 | 60.85 | <2e-16 *** |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 111 on 12262 degrees of freedom
Multiple R-squared: 0.2319, Adjusted R-squared: 0.2319
F-statistic: 3703 on 1 and 12262 DF, p-value: < 2.2e-16

- a. Which variable is my predictor variable, and which is my response variable, and how can you tell?
- 1) The predictor variable is event (whether the race is a 5k) and the response variable is pace_sec
 - 2) We know this because in the formula “lm(pace_sec ~ event, data = run_females)” the variable on the left side (pace_sec) is the response variable, and the variable on the right side (event) is the predictor variable.
- b. Write an equation for the linear model using proper notation (for the field of statistics!). Don't forget the hat! Feel free to type the word “hat” if you don't want to bother with the equation editor. If you use variables like x and y, be specific about what x and y represent in this situation.
- 1) $\text{pace_sec_hat} = 610.910 + 171.628(x)$
 - 2) Where x is a Boolean (1 or 0) for if the race is a 5k or not.
- c. Write a sentence of interpretation for each of the coefficients (slope and intercept) in the model. Write your sentences in context, rounding numbers as appropriate and using proper units.

The intercept of 610.910 represents the predicted pace in seconds for runners in the racing event (racing event being a 10k ... for example). This means that for a runner in this event, the predicted race pace is 610.91 seconds per unit distance. And the slope of 171.628 tells us how the pace changes when the event is a 5k compared to the other racing event.

What is the R^2 value? Interpret that R^2 value in context.

The R^2 value above means that 23.19% of the variability in race pace can be explained by the race event type (5k vs other races). While the remaining 76.81% of the variability is due to other factors not included in the model, such as runner fitness or training level.